ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** acetic acid	ethanoic acid
2	**(b)** basic	base test
3	**(a)** Switch connected to mains	water conducts electricity
4	**(a)** tungsten	high melting point
5	**(a)** corrosion	gradual deterioration
6	**(c)** stainless steel	alloy
7	**(c)** Cooking an egg	new substances
8	**(c)** carbon dioxide	test for CO_2
9	**(b)** iron	helps haemoglobin
10	**(b)** biodegradable pads at affordable prices	through Jan Aushadhi Kendras
11	**(c)** conduction	particles pass energy
12	**(a)** the upper level of groundwater	top of saturated zone
13	**(c)** time period	T
14	**(d)** 60 km/h	speed = distance ÷ time
15	**(c)** lungs	model
16	**(a)** down and inwards	space reduces
17	**(b)** dead and decaying matter	no chlorophyll
18	**(d)** sunlight	needed for food making
19	**(c)** almost no shadow	light passes through
20	**(a)** revolution	night sky changes

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It turns dark pink/magenta (2).	2
22	Water conducts electricity so current may pass through the body (2).	2
23	A mixture of a metal with other elements (1); steel (1).	1+1
24	Milk, curd; spinach, raisins (2).	2
25	Good: copper, aluminium; poor: wood, plastic (2).	2
26	The total distance travelled by a vehicle (2).	2
27	Tiny air sacs in the lungs (1); exchange gases with blood (1).	1+1
28	Large surface to receive light and exchange gases (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Cell (energy), bulb (light), switch (control), wires/metal strips (path), reflector (focus) (3).	3
30	Rusting is slow without flame; burning is fast with flame (3).	3
31	Heats water for evaporation and powers winds (3).	3
32	Gills cannot use air properly (3).	3
33	Look through each material at an object (1.5); classify by clarity (1.5).	1½+1½
34	Appears fixed (1.5); shows north direction (1.5).	1½+1½
35	Basic (1.5); baking soda solution (1.5).	1½+1½
	OR: Lime or slaked lime (1); it is a base and neutralises the soil acidity (2).	1+2

36	Removes green colour so iodine colour is visible (3).	3
	OR: Make food from simple substances (3).	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Salt helps corrosion (1). (ii) Paint (1). (iii) Zinc blocks (1). (iv) Sacrificial protection/galvanisation (2).	1+1+1+2
	OR: (i) Nail, wire, lead (1). (ii) Straw (1). (iii) Graphite is a conducting non-metal (1). (iv) Electrical wiring (2).	1+1+1+2
38	(i) Facial hair, voice; breast development (2). (ii) Not involved in reproduction directly (2). (iii) Reproductive organs (1).	2+2+1
	OR: (i) Diet, hygiene, exercise, sleep, social balance (3). (ii) Explanation (2).	3+2
39	(i) s = d/t (1). (ii) 80 km/h (2). (iii) 6 h (2).	1+2+2
	OR: (i) No (1). (ii) 5 m/s (1). (iii) 6 m/s (1). (iv) 5.33 m/s (2).	1+1+1+2
40	(i) Left-right reversal (2). (ii) Readable in rear-view mirror (2). (iii) Fire engine label (1).	2+2+1
	OR: (i) Long to short (2). (ii) Sun’s height changes (2). (iii) Time from shadow (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND